

Figure 3.16 The interior of the Baptistery.

I don’t know if the young Einstein had encountered the *Paradiso* during his intellectual wanderings in Italy, and whether or not the vivid imagination of the Italian poet may have had a direct influence on his intuition that the universe might be both finite and without boundary. Whether or not such influence occurred, I believe that this example demonstrates how great science and great poetry are both visionary, and may even arrive at the same intuitions. Our culture is foolish to keep science and poetry separated: they are two tools to open our eyes to the complexity and beauty of the world.

Dante’s 3-sphere is only an intuition within a dream. Einstein’s 3-sphere has mathematical form and follows from the theory’s equations. The effect of each is different. Dante moves us deeply, touching the sources of our emotions. Einstein opens a road towards the unsolved mysteries of our universe. But both count among the most beautiful and significant flights that the mind can achieve.

But let’s return to 1917, when Einstein tries to insert the idea of the 3-sphere into his equations. Here he encounters a problem. He is convinced that the universe is fixed and immutable, but his equations tell him that this is not possible. It isn’t difficult to understand why. Everything attracts, therefore the only way for a finite universe not to collapse on itself is for it to be expanding: just as the only way to prevent a football from falling to the ground is to kick it upwards. It either goes up, or falls down – it can’t stay still, suspended in the air.

But Einstein does not believe what his own equations are telling him. He even makes a silly physics mistake (he does not realize that the solution he considers is unstable) just to avoid accepting what his theory predicts: the universe is either contracting or expanding. He modifies his equations, trying to avoid the implication that it is expanding. It is for this reason that he adds